

EQUATIONAL COMPLETENESS OF  
ABSTRACT ALGEBRAS

THESIS

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William H. DeRoche, B.A.

(Washington, D.C.)

San Marcos, Texas

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William H. DeRoche

San Marcos, Texas

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## INTRODUCTION

Universal algebra is the study of classes of similar algebras rather than specific algebras. An important part of universal algebra is called model theory. A model of a set of axioms is any algebra (i.e., algebraic structure) of the appropriate type in which every axiom in the given set is valid. Each set of axioms has associated with it, then, the set of algebras that are models of the set of axioms. It is basic to this paper to understand that enlarging the set of axioms will in general reduce the set of models. For example, adding the commutative law to the set of group axioms reduces the set of models from all groups to all abelian groups. We shall be engaged in determining, in a limited context, the limits of this specialization process, that is, the point at which a set of equations has a non-trivial model, but at which stage addition as an axiom of any equation not deducible from the given ones will reduce the set of models to only the trivial one-element algebras that are models of every equation. Such a set of equations will be called "equationally complete." The term will also be applied to any non-trivial algebra that is a model of such a set of equations.

There will appear throughout this paper two types of algebraic expressions, between which the reader must

distinguish. Some will be formulas involving elements and operations of a specific algebra, in what might be called the mathematical language; and others will involve only unspecified variables and functional variables of what might be called the metamathematical language. The distinction will be carefully maintained at first but will be sacrificed for simplicity of notation later, at which point the reader should be familiar enough with the material not to become confused by the ambiguity.

The important results of the paper can be found in Kalicki and Scott [5], Kalicki [4], and Grätzer [2]. But these presentations presume a familiarity with the methods of universal algebra. This paper assumes only a knowledge of elementary abstract algebra. The books by Mac Lane and Birkhoff [7] or Herstein [3] would be more than sufficient.

## CHAPTER I

### DEFINITIONS AND FUNDAMENTAL THEOREMS

In this chapter the basic concept of equational completeness is defined and its main properties are established. The presentation is that of Kalicki and Scott [5]. For simplicity of notation, the language is restricted to that of algebras with a single binary operation.

Definition 1.1. A variable is one of the symbols "x", "y", "z", or "w", or one of those with a positive integer subscript. A functional variable is a symbol representing an operation, such as "+".

Definition 1.2. The class of terms is the least class,  $T$ , of finite sequences of variables and functional variables such that:

- (1) all variables are in  $T$ ;
- (2) if  $\alpha$  and  $\beta$  are in  $T$ , then so is  $(\alpha+\beta)$ .

Definition 1.3. The class of equations is the class of expressions  $(\alpha=\beta)$  where  $\alpha$  and  $\beta$  are terms.

Definition 1.4. If  $x$  is a variable and  $\beta$  is a term and  $\gamma$  is an equation, then  $\gamma(x|\beta)$  is the result of substituting  $\beta$  for  $x$  throughout  $\gamma$ . If  $\alpha$  is a term, we define  $\alpha(x|\beta)$  similarly.

Definition 1.5. A class  $X$  of equations is closed if:

- (1)  $(x=x)$  is in  $X$ ;
- (2) if  $(\alpha=\beta)$  and  $(\beta=\gamma)$  are in  $X$  and  $\delta$  is a term, then

$\langle \alpha = \gamma \rangle$ ,  $\langle \beta = \alpha \rangle$ ,  $\langle (\alpha + \delta) = (\beta + \delta) \rangle$ ,  $\langle (\delta + \alpha) = (\delta + \beta) \rangle$  are in  $X$ ;

(3) if  $\gamma$  is in  $X$ ,  $\beta$  is a term, and  $x$  is a variable, then  $\gamma(x|\beta)$  is in  $X$ .

Definition 1.6. The closure of a non-empty class of equations is the intersection of all closed classes of equations containing the given class. Since the class of all equations is closed, it is clear that the closure of any non-empty class exists.

Lemma 1.7. The intersection of any non-empty collection of closed classes of equations is again closed.

Proof. Let  $\{X_i\}_{i \in \Omega}$  be a non-empty collection of closed classes of equations. We want to show that  $\bigcap_{i \in \Omega} X_i$  is closed.

Since for all  $i \in \Omega$ ,  $\langle x = x \rangle \in X_i$ , then  $\langle x = x \rangle \in \bigcap_{i \in \Omega} X_i$ . Let

$\langle \alpha = \beta \rangle$  and  $\langle \beta = \gamma \rangle$  be in  $\bigcap_{i \in \Omega} X_i$  and let  $\delta$  be a term. Then for

$i \in \Omega$ ,  $\langle \alpha = \gamma \rangle$  and  $\langle \beta = \alpha \rangle$  are in  $X_i$ , since  $X_i$  is closed; and similarly,  $\langle (\alpha + \delta) = (\beta + \delta) \rangle$  and  $\langle (\delta + \alpha) = (\delta + \beta) \rangle$  are in  $X_i$ .

Since  $i$  was selected at random,  $\langle (\alpha + \delta) = (\beta + \delta) \rangle$ ,  $\langle (\delta + \alpha) = (\delta + \beta) \rangle$ ,  $\langle \alpha = \gamma \rangle$ , and  $\langle \beta = \alpha \rangle$  are in  $\bigcap_{i \in \Omega} X_i$ . Let  $\gamma$  be in  $\bigcap_{i \in \Omega} X_i$

and let  $\beta$  be a term and  $x$  a variable. Let  $i \in \Omega$ . Then  $\gamma \in X_i$ , and since  $X_i$  is closed,  $\gamma(x|\beta)$  is in  $X_i$ . Since  $i$  was selected at random,  $\gamma(x|\beta)$  is in  $\bigcap_{i \in \Omega} X_i$ . #

Lemma 1.8. The closure of a class  $X$  is the least closed class containing  $X$ .

Proof. Let  $\{X_i\}_{i \in \Omega}$  be the collection of closed sets which contain  $X$ . As noted before, this collection is non-empty. Let  $X^*$  be a closed set that contains  $X$ . Then  $X^* = X_{i_1}$  for some  $i_1 \in \Omega$ . Then  $\bigcap_{i \in \Omega} X_i \subseteq X^*$  and the closure of  $X$  is smaller than any other closed class which contains  $X$ . #

Definition 1.9. A class of equations is consistent if its closure does not contain all equations. To show a closed class of equations is consistent it is sufficient to show there exists an equation which is not an element of the class.

Remark 1.10. If  $\langle \alpha = \beta \rangle$  is an equation and  $\ddot{A} = \langle A, +_1 \rangle$  is an algebra, it is clear that if  $f$  is any scheme of replacement that systematically replaces each variable in  $\langle \alpha = \beta \rangle$  with a corresponding element of  $A$  and each  $+$  in  $\langle \alpha = \beta \rangle$  with a  $+_1$ , then the resulting expressions, " $f(\alpha)$ ", " $f(\beta)$ " represent elements of  $A$  and the resulting statement " $f(\alpha) = f(\beta)$ " is a statement that two elements of  $A$  are equal. We shall say that  $\langle \alpha = \beta \rangle$  holds in  $\ddot{A}$  if for each such replacement  $f$  the resulting statement " $f(\alpha) = f(\beta)$ " is true.

Definition 1.11. The set of identities of an algebra is the class of all equations holding in that algebra.

Definition 1.12. An algebra is a model of a class of equations if all the equations in the class hold in the algebra.

Definition 1.13. A class,  $K$ , of algebras is defined



by a set  $X$  of equations if  $K$  is the class of all models of  $X$ .  $K$  is equational if it is defined by some set of equations.

Remark 1.14. If  $X$  is the set of identities for an algebra  $\ddot{A}$ , then  $\ddot{A}$  is a model of  $X$ . However, if  $\ddot{A}$  is a model of  $X$ , then  $X$  need not be the set of identities of  $\ddot{A}$ .

Lemma 1.15. The set of identities of an algebra is closed.

Proof. Let  $\ddot{A} = \langle A, +_1 \rangle$  be an algebra and let  $X$  be the set of identities of  $\ddot{A}$ . We want to show that  $X$  is closed. We know that  $\langle x=x \rangle \in X$ , since  $\langle x=x \rangle$  holds in any algebra.

Suppose  $\langle \alpha=\beta \rangle$  and  $\langle \beta=\gamma \rangle$  are in  $X$ . If  $f$  is a replacement scheme, then  $f(\alpha) = f(\beta)$  and  $f(\beta) = f(\gamma)$  are true statements. But "=" is necessarily transitive on  $\ddot{A}$ , so  $f(\alpha) = f(\gamma)$  is true. Since  $f$  was chosen arbitrarily, it follows that  $\langle \alpha=\gamma \rangle$  holds in  $\ddot{A}$  and  $\langle \alpha=\gamma \rangle \in X$ .

Likewise, if  $f$  is a replacement scheme, then  $f(\alpha) = f(\beta)$ . But "=" is necessarily symmetric on  $A$  so  $f(\beta) = f(\alpha)$  is true. Since  $f$  was chosen arbitrarily, it follows that  $\langle \beta=\alpha \rangle$  holds in  $\ddot{A}$  and  $\langle \beta=\alpha \rangle \in X$ .

Suppose  $\delta$  is a term. If  $f$  is a replacement scheme, then  $f(\alpha) \in A$ ,  $f(\delta) \in A$ , and  $f(\beta) \in A$ . But  $\langle \alpha=\beta \rangle \in X$ , so  $f(\alpha) = f(\beta)$ . Therefore,  $f(\alpha) +_1 f(\delta) = f(\beta) +_1 f(\delta)$ , and  $\langle \alpha+\delta = \beta+\delta \rangle$  holds in  $\ddot{A}$ . Therefore  $\langle \alpha+\delta = \beta+\delta \rangle \in X$ .

Let  $\gamma$  be an equation  $\langle \alpha=\delta \rangle$  and let  $\gamma$  hold in  $X$ .

Let  $\beta$  be a term and  $x$  a variable. We want to show that  $\gamma(x|\beta)$  holds in  $\ddot{A}$ . Suppose  $f$  is an arbitrary replacement scheme, as was defined in Remark 1.10. Then  $f(\beta) \in \ddot{A}$ . Let  $\hat{f}(x)$  be the replacement scheme that assigns to  $x$  the value  $f(\beta)$  and to every other variable  $y$ ,  $f(y)$ . Then  $\hat{f}(\alpha) = f(\alpha(x|\beta))$  since  $\hat{f}$  assigns to  $x$  the value that  $f$  assigns to  $\beta$  and  $\hat{f}$  assigns the same values as  $f$  to the other variables. Also  $\hat{f}(\delta) = f(\delta(x|\beta))$ . But since ' $\alpha=\delta$ ' holds in  $\ddot{A}$ , then  $\hat{f}(\alpha) = \hat{f}(\delta)$ . Therefore,  $f(\alpha(x|\beta)) = f(\delta(x|\beta))$  is true. Then  $f(\gamma(x|\beta))$  is true and  $\gamma(x|\beta) \in X$ . Therefore,  $X$  is closed. #

Lemma 1.16. The set of identities of an algebra is consistent if and only if the algebra has at least two elements.

Proof. Necessity: Let  $\ddot{A} = \langle A, +_1 \rangle$  be an algebra such that  $A = \{a\}$ , and let  $X$  be the set of identities of  $\ddot{A}$ . Since  $A$  has one element,  $\ddot{A}$  is a model for ' $x=y$ '. Let  $\alpha$  and  $\beta$  be any terms. Since  $X$  is closed we can substitute  $\alpha$  for  $x$  and  $\beta$  for  $y$  and conclude that ' $\alpha=\beta$ '  $\in X$ . Hence, all equations are in  $X$ , and by the definition of consistency,  $X$  is not consistent. Therefore, if the set of identities of an algebra is consistent, then the algebra has to have more than one element.

Sufficiency: Let  $X$  be an inconsistent class of equations, and let  $\ddot{A}$  be a model of  $X$ . Then  $\ddot{A}$  is a model for ' $x = y$ '. Let  $a_1, a_2 \in A$ . Then the statement

$a_1 = a_2$  is true and  $A$  has only one element. Therefore, by contraposition, if  $A$  has more than one element, its set of identities is consistent. #

Lemma 1.17. Any one-element algebra is a model of any set of equations.

Proof. Let  $\ddot{A} = \langle A, +_1 \rangle$  be an algebra, where  $A$  has only one element. Since, by Lemma 1.16, the set of identities of  $\ddot{A}$  contains all equations,  $\ddot{A}$  is a model for any set of equations. #

Lemma 1.18. Every non-empty set of equations defines a unique non-empty class of algebras, its set of models.

Proof; Let  $X$  be a non-empty set of equations. Let  $K$  be the class of all algebras that are models of  $X$ . It is obvious that  $K$  is unique and  $K$  is non-empty since any trivial (one-element) algebra is a model of  $X$ . #

The reader should remember that there is not a one-to-one correspondence between sets of equations and classes of algebras. Two different classes of equations can define the same class of models.

Definition 1.19. A class of equations is equationally complete if it is consistent and is not properly contained in any consistent class of equations.

Lemma 1.20. Every equationally complete class of equations is closed.

Proof. Let  $X$  be an equationally complete class of

equations. By Definition 1.19,  $X$  is consistent; and by Definition 1.9, since  $X$  is consistent its closure does not contain all equations. But by Definition 1.19,  $X$  is not properly contained in any consistent class of equations. Therefore  $X$  is not properly contained in its closure and must equal its closure. Since, by Lemma 1.8, the closure of  $X$  is closed, then  $X$  is closed. #

Definition 1.21. An algebra is equationally complete if its set of identities is equationally complete.

Definition 1.22. A class of algebras is equationally complete if it is defined by some equationally complete set of equations.

Definition 1.23. A class  $K$  of algebras is generated by an algebra  $\ddot{A}$  if  $K$  is the class of all homomorphic images of subalgebras of direct powers of  $\ddot{A}$ .

Theorem 1.24. Every consistent class of equations is contained in some equationally complete class of equations.

Proof. Let  $Y$  be a consistent class of equations. Let  $S = \{X : Y \subseteq X \text{ and } X \text{ is a consistent closed class of equations}\}$ . It is clear that  $(S, \subseteq)$  forms a partially ordered set, where " $\subseteq$ " denotes set containment. Let  $S^*$  be a chain of sets in  $S$ . Let  $\ddot{R} = \bigcup S^*$ . We want to show that  $\ddot{R} \in S$ ; that is, that  $\ddot{R}$  is closed, consistent, and contains  $Y$ .

It can be easily shown that  $\ddot{R}$  is closed. For example, if  $(\alpha=\beta) \in \ddot{R}$  and  $(\beta=\delta) \in \ddot{R}$ , there exists  $X_1 \in S^*$  such that  $(\alpha=\beta) \in X_1$  and  $(\beta=\delta) \in X_1$  since  $S^*$  is a chain. Therefore,  $(\alpha=\delta) \in X_1 \subseteq \ddot{R}$ . We know  $\ddot{R}$  is consistent since if  $(x=y) \in \ddot{R}$ , then  $(x=y) \in X_1$  for some  $X_1 \in S^*$ . But since  $X_1$  is consistent,  $(x=y)$  can not be in  $X_1$ , so  $\ddot{R}$  must be consistent. Therefore, by Zorn's lemma there exists a maximal element in  $(S, \leq)$ , which is easily seen to be equationally complete. #

Definition 1.25. If  $x_1, \dots, x_n$  are the distinct variables in the term  $\alpha$ , and  $\alpha_1, \dots, \alpha_n$  are (not necessarily distinct) terms, then

$$\frac{S \begin{matrix} x_1 & \dots & x_n \\ \alpha_1 & \dots & \alpha_n \end{matrix} (\alpha)}{\quad}$$

is the result of simultaneously replacing  $x_1$  with  $\alpha_1$ ,  $x_2$  with  $\alpha_2$ ,  $\dots$ ,  $x_n$  with  $\alpha_n$  throughout  $\alpha$ .

Theorem 1.26. If  $X$  is a closed class of equations, and  $(\alpha = \beta) \in X$ , and  $x_1, \dots, x_n$  are all the distinct variables in both  $\alpha$  and  $\beta$ , and  $\alpha_1, \dots, \alpha_n$  are terms, then

$$\left( \frac{S \begin{matrix} x_1 & \dots & x_n \\ \alpha_1 & \dots & \alpha_n \end{matrix} (\alpha)}{\quad} = \frac{S \begin{matrix} x_1 & \dots & x_n \\ \alpha_1 & \dots & \alpha_n \end{matrix} (\beta)}{\quad} \right) \in X.$$

Proof. Let  $y_1, \dots, y_n$  be distinct variables not occurring in  $\alpha$  or in any of  $\alpha_1, \dots, \alpha_n$ . Let  $\delta_1$  be the equation  $\alpha(x_1|y_1) = \beta(x_1|y_1)$ ,  $\delta_2$  be  $\delta_1(x_2|y_2)$ , et cetera. Let  $\delta_n$  be  $\delta_{n-1}(x_n|y_n)$ ,  $\delta_{n+1}$  be  $\delta_n(y_1|\alpha_1)$ ,  $\delta_{n+2}$  be  $\delta_{n+1}(y_2|\alpha_2)$ ,  $\dots$ , and  $\delta_{n+n}$  be  $\delta_{n+n-1}(y_n|\alpha_n)$ . First, we know that  $\delta_1 \in X$

for  $i = 1, \dots, n+n$ , since  $\delta_1 \in X$  (since  $X$  is closed) and that implies that  $\delta_2 \in X$ , and by continuing this process we show  $\delta_{n+n} \in X$ . But  $\delta_{n+n}$  is the same as

$$(1) \quad \begin{matrix} ( & x_1 & \dots & x_n & ) \\ S & & & & \\ & \alpha_1 & \dots & \alpha_n & \end{matrix} (\alpha) = \begin{matrix} ( & x_1 & \dots & x_n & ) \\ S & & & & \\ & \alpha_1 & \dots & \alpha_n & \end{matrix} (\beta) .$$

Therefore, expression (1) is in  $X$ . #

Definition 1.27. If  $X$  is a class of equations, we define a relation,  $\sim$ , on the class of all terms,  $T$ , by:

$$\alpha \sim \beta \text{ if } (\alpha = \beta) \in \text{cl}(X)$$

for any two terms  $\alpha$  and  $\beta$ , where  $\text{cl}(X)$  denotes the closure of  $X$ .

Theorem 1.28. The relation defined in Definition 1.27 is an equivalence relation on the class of all terms.

Proof. We know if  $\alpha$  is a term then  $\alpha \sim \alpha$  since  $(\alpha = \alpha) \in \text{cl}(X)$ . Likewise, if  $\alpha \sim \beta$ , then  $(\alpha = \beta) \in \text{cl}(X)$  and  $(\beta = \alpha) \in \text{cl}(X)$  by the definition of "closed". Therefore,  $\beta \sim \alpha$ . Also if  $\alpha \sim \beta$  and  $\beta \sim \gamma$ , then  $(\alpha = \beta) \in \text{cl}(X)$ ,  $(\beta = \gamma) \in \text{cl}(X)$  and  $(\alpha = \gamma) \in \text{cl}(X)$  by the definition of "closed". Therefore  $\alpha \sim \gamma$  and " $\sim$ " is an equivalence relation on  $T$ . #

Definition 1.29. Let  $E(\alpha)$  be defined for  $\alpha$  a term, as  $\{\beta : \beta \text{ is a term and } (\alpha = \beta) \in \text{cl}(X)\}$ . Let  $+_1$  be defined for  $E(\alpha)$  and  $E(\beta)$  as  $E(\alpha) +_1 E(\beta) = E(\alpha + \beta)$ .

Theorem 1.30. For any non-empty class of equations, Definition 1.29 defines a binary operation on  $T/\sim$ , the set of all equivalence classes  $E(\alpha)$  for  $\alpha \in T$ .

Proof. We know  $T/\sim$  is well defined by Theorem 1.28. We also know that if  $\alpha, \beta \in T$ , then  $\alpha + \beta \in T$ . Let  $E(\alpha) = E(\beta)$  and  $E(\delta) = E(\gamma)$  for  $\alpha, \beta, \delta, \gamma \in T$ . Then  $E(\alpha) +_1 E(\delta) = E(\alpha + \delta) = \{\pi : \pi \in T \text{ and } \langle \alpha + \delta = \pi \rangle \in \text{cl}(X)\}$ . But  $\langle \alpha + \delta = \beta + \gamma \rangle \in \text{cl}(X)$  since  $\langle \alpha = \beta \rangle \in \text{cl}(X)$  and  $\langle \delta = \gamma \rangle \in \text{cl}(X)$ . So  $\langle \alpha + \delta \rangle \sim \langle \beta + \gamma \rangle$  and  $E(\alpha + \delta) = E(\beta + \gamma)$ . It follows that  $E(\alpha) +_1 E(\delta) = E(\alpha + \delta) = E(\beta + \gamma) = E(\beta) +_1 E(\gamma)$ , and  $+_1$  is a binary operation on  $T/\sim$ . #

Lemma 1.31. If  $\alpha$  is a term and  $x_1, \dots, x_n$  are distinct variables including all those in  $\alpha$ , and  $\alpha_1, \dots, \alpha_n$  are terms, and  $f$  is a replacement scheme that replaces  $x_1$  with  $E(\alpha_1)$ ,  $\dots$ ,  $x_n$  with  $E(\alpha_n)$ , and  $+$  with  $+_1$ , then  $f(\alpha) =$

$$E \left( S \begin{matrix} x_1 & \dots & x_n \\ \alpha_1 & \dots & \alpha_n \end{matrix} (\alpha) \right).$$

Proof. Let  $T$  be the class of terms as defined in Definition 1.2. Let  $W = \{\pi : \pi \text{ is a term and } f(\pi) =$

$$E \left( S \begin{matrix} x_1 & \dots & x_n \\ \alpha_1 & \dots & \alpha_n \end{matrix} (\pi) \right) \}.$$

(We assume that  $x_1, \dots, x_n$  always satisfy the hypotheses of the theorem.) We know that  $W \subseteq T$ . We want to show that  $T \subseteq W$ . To show this it is sufficient to show that all variables are in  $W$  and, if  $\alpha$  and  $\beta$  are in  $W$ , then  $\langle \alpha + \beta \rangle$  is in  $W$  (see Definition 1.2).

If  $\alpha$  is a variable, say  $x_i$  for some  $i = 1, \dots, n$ ,

then  $f(\alpha) = E(\alpha_1) =$

$$E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\alpha) \right)$$

since

$$S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\alpha) = \alpha_1.$$

Therefore,  $\alpha \in W$ , whenever  $\alpha$  is a variable.

Assume  $\beta$  and  $\gamma$  are terms and  $\beta, \gamma \in W$ . We want to show that  $\beta + \gamma \in W$ . We know that  $f(\beta) =$

$$E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\beta) \right)$$

and that  $f(\gamma) =$

$$E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\gamma) \right).$$

Suppose now that  $x_1, \dots, x_n$  is a list of distinct variables including all those in  $\beta$  or in  $\gamma$ . It is clear that  $f(\beta + \gamma)$

$$= f(\beta) +_1 f(\gamma)$$

$$= E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\beta) \right) +_1 E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\gamma) \right)$$

(by hypothesis)

$$= E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\beta) + S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\gamma) \right)$$

(by Definition 1.29)

$$= E \left( S \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} (\beta + \gamma) \right).$$



Therefore,  $\beta + \gamma \in W$ , and  $T = W$ . #

Theorem 1.32. Every consistent class of equations has a model with at least two elements.

Proof. Let  $X$  be a consistent class of equations. Then  $T = \langle T/\sim, +_1 \rangle$ , as defined before, forms an algebra. We know  $T/\sim$  has more than one element since  $X$  is consistent. We want to show that  $T$  is a model for all equations in  $X$ . Let  $\gamma \in X$  where  $\gamma$  is  $(\alpha = \beta)$  and  $\gamma$  contains the variables  $x_1, \dots, x_n$ . Let  $f$  be a replacement scheme replacing  $x_1$  with  $E(\alpha_1)$ ,  $\dots$ ,  $x_n$  with  $E(\alpha_n)$ . Then by Lemma 1.31,

$$f(\alpha) = E \left( \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ S & & & & (\alpha) \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} \right)$$

and  $f(\beta) =$

$$E \left( \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ S & & & & (\beta) \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} \right).$$

Therefore, by Theorem 1.26,

$$E \left( \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ S & & & & (\alpha) \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} \right) = E \left( \begin{array}{cccc} x_1 & \cdot & \cdot & \cdot & x_n \\ S & & & & (\beta) \\ \alpha_1 & \cdot & \cdot & \cdot & \alpha_n \end{array} \right)$$

and  $f(\alpha) = f(\beta)$ ; and  $f(\gamma)$  is true. Since  $f$  and  $\gamma$  were arbitrarily chosen,  $T$  is a model of  $X$ . #

Theorem 1.33. Every model with at least two elements of an equationally complete set  $X$  of equations is equationally complete and has  $X$  as its set of identities.

Proof. Let  $\ddot{A} = \langle A, +_1 \rangle$  be a model of  $X$ , and let  $A$  have at least two elements. Let  $Y$  be the set of identities

of  $\ddot{A}$ . Then  $X \subseteq Y$ , since all equations in  $X$  hold in  $\ddot{A}$  and  $Y$  is the set of all equations which hold in  $\ddot{A}$ . We know  $Y$  is consistent by Lemma 1.16. But, by Definition 1.19,  $X$  can not be properly contained in  $Y$  since  $X$  is maximal with respect to being consistent. Therefore,  $X = Y$  and  $\ddot{A}$  is equationally complete with  $X$  as its set of identities. #

Theorem 1.34. Every class of algebras defined by a consistent class of equations has an equationally complete element.

Proof. Let  $K$  be a class of algebras defined by  $Y$ , a consistent class of equations. By Theorem 1.24, there exists an equationally complete class of equations  $X$  such that  $Y \subseteq X$ . By Theorem 1.25, there exists a model of  $X$ ,  $\ddot{A} = \langle A, +_1 \rangle$  with at least two elements. But all the equations in  $Y$  are in  $X$  and all the equations in  $X$  hold in  $\ddot{A}$ . Therefore, all the equations in  $Y$  hold in  $\ddot{A}$  and  $\ddot{A} \in K$ . Hence  $\ddot{A}$  is equationally complete since by Theorem 1.32, a model of an equationally complete set  $X$  of equations is equationally complete. Therefore  $K$  has an equationally complete element. #

For the next theorem we will need two well known lemmas which will be stated without proof.

Lemma 1.35. (Löwenheim-Skolem) For every algebra  $\ddot{A}$  there is an at most denumerable algebra  $\ddot{A}^*$  whose identities are precisely those of  $\ddot{A}$ .

Lemma 1.36. (Birkhoff [1]) A class  $K$  of algebras is equational if and only if  $K$  is closed, that is, if and only if  $K$  contains the class generated by each element of  $K$ .

Theorem 1.37. A class of algebras is equational if and only if it is generated by some algebra.

Proof. Sufficiency: Let  $K$  be a class of algebras generated by an algebra  $\ddot{A}$ . Let  $\ddot{B} \in K$  and  $\ddot{B}^*$  be generated by  $\ddot{B}$ . It is not hard to see that  $\ddot{B}^*$  is generated by  $\ddot{A}$ , and hence  $\ddot{B}^*$  is in  $K$ . Then  $K$  contains the class generated by each element and is equational by Lemma 1.36.

Necessity: Let  $K$  be the class of all models of some set,  $X$ , of equations. Since every model of  $X$  is a model of  $\text{cl}(X)$ , we may assume without loss of generality that  $X$  is closed. Let  $\ddot{A}^*$  be the direct product of all countable algebras,  $\ddot{B}$ , in  $K$ . The existence of such algebras  $\ddot{B}$  follows from Lemma 1.35.

Since an identity holding in every factor obviously holds in the product,  $\ddot{A}^* \in K$ . So every  $\gamma \in X$  is an identity of  $\ddot{A}^*$ . Suppose  $\gamma$  is an equation and  $\gamma \notin X$ . Using the technique of the proof of 1.32 we can show that there is an algebra  $\ddot{C} \in K$  such that  $\gamma$  does not hold in  $\ddot{C}$ . (If  $f$  replaces  $x_1$  with  $E(x_1)$ ,  $\dots$ ,  $x_n$  with  $E(x_n)$  for the variables  $x_1, \dots, x_n$  in some equation  $(\alpha=\beta)$  that is not in  $\text{cl}(X)$ , then  $f(\alpha) = E(\alpha) \neq E(\beta) = f(\beta)$ , so  $(\alpha=\beta)$  does not hold in the algebra  $\langle T/\sim, +_1 \rangle$  defined in 1.29.) But by Lemma 1.35, there is a

countable algebra  $\ddot{B} \in K$  such that the identities of  $\ddot{B}$  are those of  $\ddot{C}$ . Since  $\gamma$  does not hold in  $\ddot{C}$ , it follows that  $\gamma$  does not hold in  $\ddot{B}$ . Thus  $\gamma$  can not hold in  $\ddot{A}^*$ . (The equation won't hold for the  $\ddot{B}^{\text{th}}$  coordinate in  $\ddot{A}^*$ .) Thus, the set of identities for  $\ddot{A}^*$  is  $X$ .

If  $K^*$  is the class of algebras generated by  $\ddot{A}^*$ , then  $K^* \subseteq K$  by Lemma 1.36. If  $W$  is the set of equations that hold in every element of  $K^*$  then, by 1.36,  $K^*$  is equational and must be defined by  $W$ . But  $\ddot{A}^* \in K^*$ , so  $W \subseteq X$ .

Now, if  $\ddot{A} \in K$ , then every equation in  $X$  holds in  $\ddot{A}$ , so  $\ddot{A} \in K^*$  since  $W \subseteq X$ . So  $K \subseteq K^*$  and  $K = K^*$ . Thus  $K$  is generated by an algebra. #

Theorem 1.38. If the algebra  $\ddot{A}$  is equationally complete, then any algebra with at least two elements in the class generated by  $\ddot{A}$  is equationally complete and has the same set of identities as  $\ddot{A}$ .

Proof. Let  $X$  be the set of identities of  $\ddot{A}$ . Let  $\ddot{A}^*$  be a non-trivial algebra generated by  $\ddot{A}$ . Let  $Y$  be the set of identities of  $\ddot{A}^*$ . Let  $(\alpha=\beta)$  hold in  $\ddot{A}$ . Then, since  $\ddot{A}$  generates  $\ddot{A}^*$ ,  $(\alpha=\beta)$  holds in  $\ddot{A}^*$ . Therefore, if  $(\alpha=\beta) \in X$ , then  $(\alpha=\beta) \in Y$ , and  $X \subseteq Y$ .

Since  $\ddot{A}^*$  has at least two elements,  $Y$  is consistent. By the definition of equational completeness,  $X$  can not be properly contained in  $Y$ . Hence  $X = Y$ . Therefore,  $\ddot{A}^*$  is equationally complete and has the same set of identities as  $\ddot{A}$ . #

Theorem 1.39. An equational class of algebras is equationally complete if and only if it is generated by some equationally complete algebra.

Proof. Necessity: Let  $K$  be an equationally complete class of algebras defined by  $X$ , an equationally complete set of equations. That is, each algebra in  $K$  is a model of  $X$  and all models of  $X$  are in  $K$ .

By Theorem 1.37,  $K$  is generated by some algebra, say the algebra  $\ddot{A} = \langle A, +_1 \rangle$ . If  $A$  had only one element, every element of  $K$  would be a model for every equation and  $K$  would have no equationally complete element, a contradiction of Theorem 1.34. So  $A$  has at least two elements. But  $\ddot{A}$  is in  $K$  since  $\ddot{A}$  is a homomorphic image of itself. Hence,  $\ddot{A}$  is a model of  $X$ . Therefore, by Theorem 1.33,  $\ddot{A}$  is an equationally complete algebra that generates  $K$ .

Sufficiency: Let  $K$  be a class of algebras generated by  $\ddot{A}$ , an equationally complete algebra. Let  $X$  be the set of identities of  $\ddot{A}$ . We wish to show that  $K$  is defined by  $X$ , that is, that  $K$  is the class of all models of  $X$ .

Since  $K$  is generated by  $\ddot{A}$ , it is clear that  $K$  is closed. By Lemma 1.36,  $K$  is equational. Suppose  $K$  is defined by the class of equations  $W$ . Then  $W \subseteq X$  since  $\ddot{A} \in K$ . As seen in the proof of Theorem 1.37, there is an algebra  $\ddot{B} \in K$  such that the set of identities of  $\ddot{B}$  is exactly  $\text{cl}(W)$ .

Since  $X$  is consistent and  $X \supseteq W$ ,  $\text{cl}(W)$  is consistent and  $B$  has at least two elements (by Lemma 1.16). By Theorem 1.38, the set of identities of  $\ddot{B}$  is  $X$ . So  $X = \text{cl}(W)$  and  $K$  is the set of all models of  $X$ . So  $K$  is equationally complete. #

The following are some related theorems of Tarski:

Theorem 1.40. The class of algebras generated by an algebra  $\ddot{A}$  coincides with the class of all algebras in which every identity of  $\ddot{A}$  holds.

Corollary 1.41. For any two algebras,  $\ddot{A}$  and  $\ddot{B}$ , the following are equivalent:

- (i)  $\ddot{A}$  and  $\ddot{B}$  have the same identities;
- (ii) Each of  $\ddot{A}$  and  $\ddot{B}$  belong to the class generated by the other;
- (iii) The class of algebras generated by  $\ddot{B}$  coincides with the class of algebras generated by  $\ddot{A}$ .

Theorem 1.42. For any equational class of algebras,  $K$ , in which at least one member has at least two elements, the following three conditions are equivalent:

- (i)  $K$  is equationally complete;
- (ii) Every algebra in  $K$  with more than one element is equationally complete;
- (iii) Any two algebras in  $K$  having more than one element have the same set of identities.

The proofs of these three results are similar to the preceding ones and may be found in Tarski [8].

## CHAPTER II

### EQUATIONAL COMPLETENESS OF ASSOCIATIVE ALGEBRAS

In this section we will give necessary and sufficient conditions for an associative algebra to be equationally complete, thus determining the practical limits of specialization in the class of associative algebras.

Definition 2.1. If  $\ddot{A} = \langle A, + \rangle$  is an algebra and  $S$  is a non-empty subset of  $A$ , then  $\ddot{A}[S]$  denotes the subalgebra generated by  $S$ . That is,  $\ddot{A}[S]$  is the smallest subalgebra of  $\ddot{A}$  that contains  $S$ .

Definition 2.2. We make the following abbreviations, where  $\alpha$  is a term and  $n$  is a positive integer:

$$\begin{aligned} & ' [1\alpha] ' \text{ for } \alpha \\ & ' [(n+1)\alpha] ' \text{ for } ' ([n\alpha] + \alpha) ' . \end{aligned}$$

Definition 2.3. The class of associative algebras (AS) is the class of all algebras in which  $'(x+(y+z)) = ((x+y)+z)'$  holds.

Definition 2.4. The class of semi-lattices (SL) is the class of all algebras in AS in which both  $'(x+y) = (y+x)'$  and  $'(x+x) = x'$  hold.

Definition 2.5. The class of left algebras (LF) is the class of all algebras in AS in which  $'(x+y) = x'$  holds.

Definition 2.6. The class of right algebras (RG) is the class of all algebras in AS in which  $\langle (x+y) = y \rangle$  holds.

Definition 2.7. The class of constant algebras (CS) is the class of algebras in AS in which  $\langle (x+y) = (z+w) \rangle$  holds.

Remark 2.8. Clearly, SL, LF, RG, and CS are all included in AS.

Definition 2.9. If  $p$  is a prime ( $p \geq 2$ ), then the class of elementary p-groups (p-GR) is the class of all algebras in AS for which both  $\langle (x+y) = (y+x) \rangle$  and  $\langle ([px]+y) = y \rangle$  hold.

Remark 2.10. Each element of p-GR is necessarily a "primary abelian p-group," that is, a group in which every non-identity element has order  $p$ . If  $\ddot{A} \in \text{p-GR}$ , the equation  $\langle px = py \rangle$  is easily shown to hold in  $\ddot{A}$ . Since  $\langle px+y = y \rangle$  holds, there is a left identity element. And since  $\langle [(p-1)x] + x = px \rangle$  holds, every element has a left inverse. So each  $\ddot{A} \in \text{p-GR}$  is a primary abelian p-group.

Definition 2.11.  $K$  is in CM if  $K$  is in SL, LF, RG, CS, or p-GR for some prime  $p$ .

Remark 2.12. It is clear from the way they are defined that each of the classes in CM is equational.

Definition 2.13. If  $K$  is in CM, then we denote by  $\ddot{C}(K)$  the characteristic algebra of  $K$ , which is defined as follows:

- (1)  $\ddot{C}(\text{SL}) = \langle \{0,1\}, +_2 \rangle$ , where the operation  $+_2$  is defined by the addition table:



| $+_2$ | 0 | 1 |
|-------|---|---|
| 0     | 0 | 0 |
| 1     | 0 | 1 |

(2)  $\ddot{C}(LF) = \langle \{0,1\}, +_3 \rangle$ , where  $+_3$  is defined by:

| $+_3$ | 0 | 1 |
|-------|---|---|
| 0     | 0 | 0 |
| 1     | 1 | 1 |

(3)  $\ddot{C}(RG) = \langle \{0,1\}, +_4 \rangle$ , where  $+_4$  is defined by:

| $+_4$ | 0 | 1 |
|-------|---|---|
| 0     | 0 | 1 |
| 1     | 0 | 1 |

(4)  $\ddot{C}(CS) = \langle \{0,1\}, +_5 \rangle$ , where  $+_5$  is defined by:

| $+_5$ | 0 | 1 |
|-------|---|---|
| 0     | 0 | 0 |
| 1     | 0 | 0 |

(5) If  $p$  is a prime, then  $\ddot{C}(p\text{-GR}) = \langle \{0,1,\dots,p-1\}, +_p \rangle$ , where  $+_p$  is the operation of addition modulo  $p$ .

Remark 2.13. It is obvious that if  $K$  is in CM then

$\ddot{C}(K)$  is in K.

Theorem 2.14. Every class in CM is generated by its

characteristic algebra. This means, for example, that if  $\ddot{A}$  is any semi-lattice, then  $\ddot{A}$  is isomorphic to a member of the class generated by  $\ddot{C}$  (SL).

Proof. Let  $\ddot{A} = \langle A, +_0 \rangle$  be in CM.

Case (1):  $\ddot{A} = \langle A, +_0 \rangle$  is in SL. Let  $S$  be the set of all subsets of  $A$ . For  $a$  in  $A$  define  $F(a)$  as the set of all  $b$  in  $A$  such that  $a +_0 b = b$ . We want to show that  $\ddot{A}$  is mapped by  $F$  isomorphically onto a sub-semi-lattice of the algebra  $\langle S, \cap \rangle$ .

Let  $a \in A$ ; then  $F(a)$  is a subset of  $A$  and  $F(a) \in S$ . Let  $a = a^* \in A$ . If  $b \in F(a^*)$ , then  $a^* +_0 b = b$ . Since  $a = a^*$ ,  $a +_0 b = b$  and  $b \in F(a)$ . Therefore  $F(a^*) \subseteq F(a)$ . By a similar proof,  $F(a) \subseteq F(a^*)$ . Therefore,  $F(a) = F(a^*)$ , and  $F$  is well defined.

Let  $a$  and  $c$  be in  $A$ . Let  $b \in F(a) \cap F(c)$ . Then  $a +_0 b = b$  and  $c +_0 b = b$ . But  $(a +_0 c) +_0 b$

$$\begin{aligned}
 &= a +_0 (c +_0 b) && \text{(since } +_0 \text{ is associative)} \\
 &= a +_0 b && \text{(since } c +_0 b = b) \\
 &= b && \text{(since } a +_0 b = b).
 \end{aligned}$$

Therefore,  $b \in F(a +_0 c)$ , and  $F(a) \cap F(c) \subseteq F(a +_0 c)$ .

Now, let  $b \in F(a +_0 c)$ . Then  $(a +_0 c) +_0 b = b$ . But  $(a +_0 b)$

$$\begin{aligned}
 &= a +_0 ((a +_0 c) +_0 b) \\
 &\hspace{15em} \text{(since } (a +_0 c) +_0 b = b) \\
 &= ((a +_0 a) +_0 c) +_0 b
 \end{aligned}$$

(by associativity of  $+_0$ )

$$= (a +_0 c) +_0 b = b.$$

Therefore,  $b \in F(a)$ . By a similar argument,  $b \in F(c)$ .

Therefore,  $b \in F(a) \cap F(c)$  and  $F(a +_0 b) \subseteq F(a) \cap F(c)$ .

Therefore,  $F(a +_0 b) = F(a) \cap F(c)$  and  $F$  is a homomorphism mapping  $\ddot{A}$  into  $\langle S, \cap \rangle$ .

Let  $F(a) = F(c)$  for some  $a, c \in A$ . Then  $a \in F(a)$ , since  $a +_0 a = a$ . Likewise,  $c \in F(c)$ , since  $c +_0 c = c$ . But  $a \in F(c)$ , since  $F(a) = F(c)$ , so  $c +_0 a = a$ . Also  $c \in F(a)$ , so  $a +_0 c = c$ . But  $c +_0 a = a +_0 c$ , so  $a = c$ . Thus,  $F$  is one-to-one; and  $F$  is an isomorphism onto a subalgebra of  $\langle S, \cap \rangle$ .

Now we wish to show that  $\langle S, \cap \rangle$  is isomorphic to the direct product of the algebra  $\ddot{C}(SL)$  with index set  $A$ . Let  $\ddot{B} = \prod_{a \in A} \ddot{C}(SL)_{a_1}$ . Pick  $K \in \ddot{B}$  and let  $K_a$  denote the  $a^{\text{th}}$  coordinate of  $K$  for all  $a \in A$ . Notice that  $K_a = 0$  or  $1$  for each  $a \in A$ . Then define a map

$$g : \ddot{B} \longrightarrow \langle S, \cap \rangle,$$

such that for all  $K \in \ddot{B}$ ,  $g(K) = \{a : K_a = 1\}$ .

We want to show that  $g$  is well defined. Let  $K = K^* \in \ddot{B}$ . Then by the definition of product set, for every  $a \in A$ ,  $K_a = K_a^*$ . It follows that  $K_a = 1$  if and only if  $K_a^* = 1$ ; so  $g(K) = g(K^*)$ .

To show that  $g$  is a homomorphism, let  $K$  and  $K^*$  be arbitrary elements of  $\ddot{B}$ . Then  $a \in g(K + K^*)$  if and only

if  $(K + K^*)_a = 1$ , which occurs if and only if  $K_a + K_a^* = 1$ . The table shows that  $K_a + K_a^* = 1$  if and only if  $K_a = 1$  and  $K_a^* = 1$ . This is true if and only if  $a \in g(K)$  and  $a \in g(K^*)$ . Therefore  $g(K + K^*) = g(K) \cap g(K^*)$ .

Next we wish to show that  $g$  is one-to-one. Let  $g(K) = g(K^*)$  for some  $K$  and  $K^*$ . Let  $K_{a_1}$  be the  $a_1^{\text{th}}$  coordinate of  $K$ . Then  $K_{a_1} = 1$  implies that  $a_1 \in g(K)$ , and  $a_1 \in g(K^*)$ , and 1 is also the  $a_1^{\text{th}}$  coordinate of  $K$ . This is also true if  $K_{a_1} = 0$ . Therefore  $K = K^*$  and  $g$  is one-to-one. Therefore, an isomorphic copy of  $\ddot{A}$  is in the class generated by  $\ddot{C}$  (SL). At this point, where it is unlikely that any confusion will arise for perceptive readers, the numerical subscripts on operations and elements will not be so carefully used to distinguish between metamathematical symbols. For example, in a left algebra "+" will be used for "+<sub>3</sub>" to make the text more legible. To further simplify the text, small letter "a's" with subscripts will denote arbitrary elements of  $A$ .

Case (2): Let  $\ddot{A} \in \text{LF}$ . Let  $\ddot{B} = \prod_{a \in A} \ddot{C}(\text{LF})_a$ ,

the direct product of characteristic left algebras with index set  $A$ . Let  $g : \ddot{A} \rightarrow \ddot{B}$  be defined such that, for every  $a, b \in A$ ,  $g(a)_b = 1$  if and only if  $a = b$ , where  $g(a)_b$  denotes the  $b^{\text{th}}$  coordinate of  $g(a)$ . We know  $g$  is well defined since if  $a = a^*$ , then  $g(a)_b = g(a^*)_b$  for all  $b \in A$ , so  $g(a) =$

$g(a^*)$ . We know  $g$  is a homomorphism since  $g(a) + g(a^*) = g(a^*) = g(a + a^*)$ , since  $\ddot{A}$  and  $\ddot{B}$  are both left algebras.

Finally,  $g$  is an isomorphism since, if  $g(a) = g(a^*)$ , then for all  $c \in A$ ,  $g(a)_c = g(a^*)_c$ . Therefore,  $g(a^*)_a = 1$  and  $a = a^*$ , by the definition of  $g$ . Therefore  $\ddot{A}$  is isomorphic to a subalgebra of  $\ddot{B}$ , since it is clear that  $g(\ddot{A})$  is a subalgebra of  $\ddot{B}$ .

Case (3):  $\ddot{A} \in \text{RG}$ . The proof is essentially the same as in case (2).

Case (4): Let  $\ddot{A} \in \text{CS}$ . Let  $\ddot{B} = \prod_{a \in A} \ddot{C}(\text{CS})_a$ .

Since  $\ddot{A} \in \text{CS}$ , there is a special element,  $0$ , in  $A$  such that for all  $a, a^* \in A$ ,  $a + a^* = 0$ . We will define  $g: \ddot{A} \rightarrow \ddot{B}$  as in case (2). We know  $g$  is well defined. We also see that  $g(0)$  is the constant element of  $\ddot{B}$ . Then  $g(a + a^*) = g(0) = g(a) + g(a^*)$ , and  $g$  is a homomorphism. We know  $g$  is one-to-one since  $g$  was defined the same as in case (2). So  $g$  is an isomorphism of  $\ddot{A}$  onto a subalgebra of  $\ddot{B}$ .

Case (5): Let  $\ddot{A}$  be in  $p\text{-GR}$ . Let  $\ddot{B} = \prod_{a \in A} \ddot{C}(p\text{-GR})_a$ ,

the product algebra generated by  $\ddot{C}(p\text{-GR})$ . If  $f \in \ddot{B}$ , we say  $f \in \ddot{F}$  if  $\{a : a \in A \text{ and } f_a \neq 0\}$  is finite. It is obvious that  $\ddot{F} \subseteq \ddot{B}$ . We know if  $f \in \ddot{F}$  and  $f^* \in \ddot{F}$ , then  $f + f^* \in \ddot{F}$  since  $\{a : a \in A \text{ and } (f + f^*)_a \neq 0\}$  is finite. So  $\ddot{F}$  is a subalgebra of  $\ddot{B}$ . We define a mapping  $\sigma$  from  $\ddot{F}$  into  $\ddot{A}$  by  $\sigma(f) = \sum_{a \in A} f_a$ . It is obvious that  $\sigma$  is well

defined. We know  $\sigma$  is a homomorphism since  $\sigma(f + f^*) = \sum_{a \in A} (f + f^*)_a = \sum_{a \in A} (f)_a + \sum_{a \in A} (f^*)_a = \sigma(f) + \sigma(f^*)$ .

Let  $a^* \in A$ . Then there exists  $f \in \ddot{F}$  such that  $f_a \neq 0$  if and only if  $a = a^*$ . But  $\sigma(a^*) = a$ . Therefore  $\sigma$  is onto. Therefore there exists a subalgebra of  $\ddot{B}$  that is isomorphic to  $\ddot{A}$  and therefore  $\ddot{A}$  is generated by its characteristic algebra. #

Theorem 2.15. If  $K$  is in CM and  $\ddot{A}$  is in  $K$  and has at least two elements, then  $\ddot{C}(K)$  is isomorphic to a subalgebra of  $\ddot{A}$ .

Proof. This proof will show that all non-trivial algebras in CM contain (that is, have a subalgebra isomorphic to) a characteristic algebra. Let  $\ddot{A} = \langle A, +_0 \rangle$  be an algebra in some class  $K$  in CM.

Case (1):  $\ddot{A}$  is in SL. Let  $a, b$  be two elements of  $A$  and let  $a \neq b$ . We want to find a sub-semi-lattice of  $\ddot{A}$  with two elements. Either  $a + b = a$  or  $a + b \neq a$ . If  $a + b = a$ , then  $a + a = a$ ,  $b + b = b$ , and  $a + b = a = b + a$ . Then  $\{a, b\}$  is closed under  $+$ . If  $a + b \neq a$ , then  $(a + b) + (a + b) = (a + b)$ , and  $a + a = a$ , and  $(a + b) + a = a + (a + b) = a + b$ . Hence  $\{a, a+b\}$  is closed under  $+$ . In either case there is a subalgebra isomorphic to the characteristic algebra  $\ddot{C}(SL)$ .

Case (2): Let  $\ddot{A} \in LF$ . Let  $a, b \in A$  and  $a \neq b$ . Then  $a + a = a$ ,  $b + a = a$ ,  $b + b = b$ , and  $a + b = b$ .

Therefore, the set  $\{a,b\}$  is closed under  $+$  and forms the desired subalgebra.

Case (3): Let  $\ddot{A} \in \text{RG}$ . The proof is similar to the proof used in case (2).

Case (4): Let  $\ddot{A} \in \text{CS}$ . Let  $a, b \in A$  and  $a \neq b$ . Then either  $a + a = a$  or  $a + a \neq a$ . If  $a + a = a$ , then  $a = a + a = a + b = b + a = b + b$  and  $\{a,b\}$  is closed under  $+$ . If  $a + a \neq a$ , then  $a + a = (a + a) + (a + a) = (a + a) + a = a + (a + a) = a + a$ . Therefore,  $\{a, (a+a)\}$  is closed under  $+$  and forms the desired subalgebra.

Case (5): Let  $\ddot{A} \in \text{p-GR}$  for  $p$  a prime. Since  $\ddot{A}$  is a group with more than one element, there exists  $x \in A$  such that  $x + x \neq x$ . Consider  $\ddot{B} = \{ix : 1 \leq i < p\}$ .  $\ddot{B}$  is a subgroup of  $\ddot{A}$  by a well known group theory result. Since  $\ddot{B}$  has  $p$  elements, it follows by elementary group theory results that  $\ddot{B}$  is a cyclic group of order  $p$ . But any two cyclic groups of order  $p$  are known to be isomorphic. Therefore,  $\ddot{B}$  is isomorphic to  $\ddot{C}$  (p-GR) and  $\ddot{A}$  contains a subalgebra which is isomorphic to  $\ddot{C}$  (p-GR). #

The next result says that all associative algebras have a subalgebra that can be mapped homomorphically onto one of the five characteristic algebras. This theorem will permit the determination of all equationally complete associative algebras.

Theorem 2.16. If  $\ddot{A}$  is an algebra in AS with at least

two elements, then, for some  $K$  in CM,  $\ddot{C}(K)$  is a homomorphic image of a subalgebra of  $\ddot{A}$ .

Proof. Let  $\ddot{A} = \langle A, + \rangle$ . Either  $'(x + x) = x'$  holds in  $\ddot{A}$  or there exists  $a \in A$  such that  $a + a \neq a$ .

Case (1): If  $'(x + x) = x'$  holds then either  $'(x + y) = (y + x)'$  holds or there exists  $a$  and  $b$  in  $A$  such that  $a + b \neq b + a$ .

Subcase (a): If  $'(x + y) = (y + x)'$  holds, then  $A$  is a semi-lattice and, by Theorem 2.15,  $\ddot{C}(SL)$  is isomorphic to a subalgebra of  $\ddot{A}$ , and hence, is a homomorphic image of a subalgebra of  $\ddot{A}$ .

Subcase (b): In case  $'(x + y) = (y + x)'$  does not hold in  $\ddot{A}$ , then there exist  $a$  and  $b$  in  $A$  such that  $a + b \neq b + a$ . Then let  $c = a + b$  and  $d = a + (b + a)$ . Then  $c$  and  $d$  are in  $A$ .

$$\begin{aligned}
 & \text{Part 1: Let } c \neq d. \text{ Then } c + d \\
 &= (a + b) + (a + (b + a)) \\
 & \hspace{15em} (\text{by definition of } c \text{ and } d) \\
 &= ((a + b) + (a + b)) + a \hspace{5em} (\text{by associativity}) \\
 &= (a + b) + a \hspace{10em} (\text{since } (a+b)+(a+b)=a+b) \\
 &= d. \hspace{15em} (\text{by associativity}).
 \end{aligned}$$

Likewise,  $d + c$

$$\begin{aligned}
 &= (a + (b + a)) + (a + b) \\
 & \hspace{15em} (\text{by definition of } d \text{ and } c) \\
 &= (a + b) + ((a + a) + b) \hspace{5em} (\text{by associativity})
 \end{aligned}$$



$$\begin{aligned}
&= (a + b) + (a + b) && \text{(since } a + a = a \text{)} \\
&= a + b && \text{(since } (a+b)+(a+b)=a+b \text{)} \\
&= c.
\end{aligned}$$

Since  $c + c = c$  and  $d + d = d$ , the set  $\{d, c\}$  is closed under  $+$  and is isomorphic to  $\check{C}$  (RG).

Part 2: Let  $c = d$ , and let  $f = b + a$ . Since  $c = a + b \neq b + a = f$ , it follows that  $c \neq f$ . Then  $c + f$

$$\begin{aligned}
&= (a + b) + (b + a) && \text{(by definition of } c \text{ and } f \text{)} \\
&= a + ((b + b) + a) && \text{(by associativity)} \\
&= a + (b + a) && \text{(since } b + b = b \text{)} \\
&= d = c.
\end{aligned}$$

So  $f + c = f + d$

$$\begin{aligned}
&= (b + a) + (a + (b + a)) \\
&&& \text{(by definition of } f \text{ and } d \text{)} \\
&= (b + (a + a)) + (b + a) && \text{(by associativity)} \\
&= (b + a) + (b + a) && \text{(since } a + a = a \text{)} \\
&= b + a && \text{(since } (b+a)+(b+a)=b+a \text{)} \\
&= f.
\end{aligned}$$

Therefore, since  $f + f = f$  and  $c + c = c$ , then  $\{c, f\}$  is isomorphic to  $\check{C}$  (LF); that is,  $\check{C}$  (LF) is isomorphic to a subalgebra of  $\check{A}$ .

Case (2): We assume that there exists some  $a \in A$  such that  $a + a \neq a$ . Let  $[\{a\}] = \{ia : i \text{ is a positive integer}\}$ .  $\check{A}[\{a\}] = \langle [\{a\}], + \rangle$ , where  $[\{a\}] \subseteq A$ .

It is clear that  $\ddot{A}[\{a\}]$  is a subalgebra of  $\ddot{A}$ . As is customary, we understand that  $1 \cdot a = a$ ,  $2 \cdot a = a + a$ , et cetera. Either  $na = ma$  implies  $m = n$  for all integers  $m, n$ , or there exist integers  $m, n$  such that  $na = ma$  and  $n \neq m$ .

Subcase (a): If  $na = ma$  implies  $m = n$ , we define a mapping,  $g$ , by  $g(2n \cdot a) = 0$  and  $g((2n+1) \cdot a) = 1$ , for all integers  $n$ . It is obvious that  $g$  is well defined and a homomorphism from  $A[\{a\}]$  onto  $\ddot{C}$  (2-GR). Therefore,  $\ddot{C}$  (2-GR) is a homomorphic image of a subalgebra of  $\ddot{A}$ .

Subcase (b): There exist  $m, n$  such that  $m \neq n$  and  $ma = na$ . Let  $n_1$  be the least positive integer such that there exists  $m \neq n_1$  for which  $n_1 a = ma$ . Let  $m_1$  be the least integer such that  $m_1 > n_1$  and  $n_1 a = m_1 a$ . Either  $m_1 = n_1 + 1$  or  $m_1 \neq n_1 + 1$ .

Part 1: Let  $m_1 = n_1 + 1$ . Then either  $n_1 = 1$  or  $n_1 \neq 1$ . But if  $n_1 = 1$ , then  $a = n_1 a = (n_1 + 1)a = 2a$  and  $a + a = a$ . This is a contradiction, since we assumed that  $a \neq a + a$ . Therefore  $n_1 \neq 1$ . Suppose  $k = \ell + n_1$ , where  $\ell > 0$ . Then  $ka = (\ell + n_1)a = ((\ell - 1) + n_1 + 1)a = (\ell - 1)a + (n_1 + 1)a$  (by associativity)  
 $= (\ell - 1)a + m_1 a = (\ell - 1)a + n_1 a = (\ell - 1 + n_1)a = (k - 1)a$ .  
 Therefore, if  $k > n_1$ ,  $ka = (k - 1)a$ . But either  $(k - 1) = n_1$  or  $(k - 1) > n_1$ . If  $(k - 1) = n_1$ , then  $ka = n_1 a$ . If  $(k - 1) > n_1$ , then  $(k - 2)a = ka$ , as before. By continuing this process, we must obtain an  $i$  such that  $ka = (k - i)a$  and  $k - i = n_1$ .

Therefore, if  $k > n_1$ , then  $ka = n_1 a$ . Consider the set  $\{(n_1-1)a, n_1 a\}$ . This set is closed under "+" since the sum of any two elements in the set is equal to  $n_1 a$ . Hence,  $\{(n_1-1)a, n_1 a\}$  is isomorphic to  $\ddot{C}$  (CS).

Part 2: Let  $m_1 > n_1 + 1$ . Consider the set  $\ddot{D} = \{n_1 a, (n_1+1)a, \dots, (m_1-1)a\}$ . By the selection of  $m_1$  and  $n_1$ , if  $ka = \ell a$ , where  $k$  and  $\ell$  are positive integers and  $k \neq \ell$ , then  $|k-\ell| \geq |m_1-n_1|$ . Suppose  $r$  is an integer and  $r \geq n_1$ . If  $r \geq m_1$ , say  $r = m_1 + t$  for some  $t \geq 0$ , then  $ra = (m_1+t)a = m_1 a + ta = n_1 a + ta = (n_1+t)a$ . Thus, if  $r \geq m_1$ , there exists an integer  $k$  such that  $ra = ka$  and  $r > k \geq n_1$ . If  $k \geq m_1$ , the same reduction is possible again. In a finite number of such steps the process must terminate with an integer  $h$  such that  $ra = ha$  and  $m_1 > h \geq n_1$ . It is obvious from this that  $\ddot{D}$  is closed under "+", since the sum of two elements of  $\ddot{D}$  is clearly expressible as  $ra$  for some  $r \geq n_1$ .

Suppose that  $ka + \ell a = sa + \ell a$ , where  $m_1 > k \geq s \geq n_1$ . Then  $k = s + g$  for some integer  $g$  such that  $g \geq 0$ . It follows that  $(s+g)a + \ell a = sa + \ell a$ , that is,  $(g+s+\ell)a = (s+\ell)a$ . But  $(g+s+\ell) - (s+\ell) = g = k - s < m_1 - n_1$ . Then  $g = 0$  and  $k = s$  because of the choice of  $m_1$  and  $n_1$ . Hence the cancellation law holds and  $\ddot{D}$  is a group with order  $m_1 - n_1$ , and is a subalgebra of  $\ddot{A}$ .

Let  $p$  be a prime factor of  $m_1 - n_1$ . Then by Cayley's theorem, there exists an  $x \in A$  with order

p. The set  $\{[x]\}$  is a group of order p. Therefore  $\ddot{A}$  contains a set which is isomorphic to  $\ddot{C}$  (p-GR) and thus  $\ddot{C}$  (p-GR) is a homomorphic image of a subalgebra of  $\ddot{A}$ .

Therefore, we have proven that all associative algebras have a subalgebra which is isomorphic to one of the five characteristic algebras. #

Theorem 2.17. If  $\ddot{A}$  is in AS and is equationally complete, then for some K in CM,  $\ddot{A}$  is in K.

Proof. By previous results, since  $\ddot{A}$  is equationally complete it has at least two elements. Then by Theorem 2.16, there is a K in CM such that  $\ddot{C}(K)$  is in the class of algebras generated by  $\ddot{A}$ . Therefore, since  $\ddot{C}(K)$  has two elements and is in the class generated by  $\ddot{A}$ ,  $\ddot{C}(K)$  has the same set of identities as  $\ddot{A}$ , by Theorem 1.38. But an algebra that has the same set of identities as  $\ddot{C}(K)$  is in K. Therefore,  $\ddot{A}$  is in K. #

Theorem 2.18. All classes in CM are equationally complete.

Proof. Let K be in CM. As noted in Remark 2.12, K is equational. By Theorem 1.27, there is an equationally complete algebra in K. By Theorem 2.15,  $\ddot{C}(K)$  is in the class generated by this algebra and hence is equationally complete. Since it was shown in Theorem 2.14 that K was generated by  $\ddot{C}(K)$ , K is equationally complete. #

Theorem 2.19. The only equationally complete classes of algebras included in AS are the classes in CM; or, in other words, an equational associative algebra is equationally complete if and only if it is a semi-lattice, a right algebra, a left algebra, a constant algebra, or a primary p-group for some prime p.

Proof. This follows directly from Theorems 2.17 and 2.18. #

## CHAPTER III

### THE NUMBER OF EQUATIONALLY COMPLETE CLASSES OF EQUATIONS

Since the requirement of equational completeness is so restrictive, the set of equationally complete classes of equations would appear to be rather small. Actually, the set has the same cardinality as that of the continuum. That is, there is a one-to-one correspondence between the set of equationally complete classes and the set of all non-empty sets of integers. It will turn out that the set of equationally complete commutative algebras has that same cardinality. This is an interesting contrast to the results of Chapter II, which show that there are only countably many equationally complete classes of associative algebras.

Convention 3.1. If  $\alpha$  is a term we shall abbreviate the string  $'(\alpha+\alpha)'$  by  $2^1\alpha$  (or simply by  $'2\alpha'$ ) and, for  $n > 0$ , we shall abbreviate  $'2(2^n\alpha)'$  by  $'2^{n+1}\alpha'$ . Similarly, if  $a$  is an element of an algebra  $\ddot{A} = \langle A, +_1 \rangle$ , we shall write  $2^1a$  (or  $2a$ ) for  $a +_1 a$  and, in case  $n > 0$  we shall write  $2^{n+1}a$  for  $2(2^n a)$ .

Convention 3.2. If  $M$  and  $N$  are two disjoint sets of positive integers, then  $X(M,N)$  will stand for the set of equations consisting of

$$(i) \quad (2x + x = 2y + y);$$

$$(ii) \quad (2^m x + x = 2x + x) \text{ for all } m \text{ in } M;$$

$$(iii) \quad (2^n x + x = x) \text{ for all } n \text{ in } N.$$

Lemma 3.3. Let  $\ddot{A} = \langle A, +_1 \rangle$  be any algebra with countably many elements,  $a_1, a_2, \dots, a_1, \dots$  and such that for all positive integers  $i$ ,  $a_1 +_1 a_i = a_{i+1}$ ; then  $2^m a_n = a_{n+m}$  for all positive integers  $m, n$ .

Proof. By induction on  $m$ . Let  $m = 1$ . Then

$$\begin{aligned} 2^m a_n &= 2a_n && (\text{since } m=1) \\ &= a_n +_1 a_n \\ &= a_{n+1} && (\text{by the assumed property of } +_1) \\ &= a_{n+m} && (\text{since } m=1). \end{aligned}$$

Assume the theorem is true for  $m-1$ . Then  $2^{m-1} a_n = 2a_{n+m-1}$ .

This implies that

$$\begin{aligned} 2^m a_n &= 2^{m-1} a_n +_1 2^{m-1} a_n \\ &= 2a_{n+m-1} +_1 2a_{n+m-1} \\ &= 2a_{n+m} && (\text{since } 2a_{n+m-1} = a_{n+m}). \end{aligned}$$

Therefore,  $2^m a_n = 2a_{n+m}$  for all positive integers  $n, m$  and all  $a \in A$ . #

Definition 3.4. Two sets of integers are called complementary if and only if they are disjoint and their union consists of all integers greater than 1. (An example would be the set of all odd integers greater than one and the set of positive even integers.)

Lemma 3.5. Let  $M$  and  $N$  be complementary sets of integers, with  $1 \in M$ . Then  $X(M, N)$  is consistent.

Proof. We will exhibit an algebra  $\ddot{A}(M,N) = \langle A, +_1 \rangle$  with more than one element in which all the equations of the set  $X(M,N)$  hold. Let  $A = \langle A, +_1 \rangle$ , where  $A = \{a_1, a_2, \dots, a_i, \dots\}$  and the operation  $+_1$  is given as follows: for every positive integer  $i$ ,

$$(i) \quad a_{i+m} +_1 a_i = a_i, \text{ for all } m \text{ in } M;$$

$$(ii) \quad a_{i+n} +_1 a_i = a_i, \text{ for all } n \text{ in } N;$$

$$(iii) \quad a_i +_1 a_i = a_{i+1}.$$

We note that  $a_i +_1 a_j$  has been defined only in the case  $j \leq i$ . This means that we are completely free in defining  $a_i +_1 a_j$  for the remaining cases. We define the remaining cases in such a way that " $+_1$ " will be commutative on  $A$ . It is not hard to show that  $\ddot{A}$  is a model of  $X(M,N)$ . Since  $A$  has more than one element,  $X(M,N)$  is consistent by Lemma 1.16. #

Lemma 3.6. If  $M$  and  $N$  are complementary sets of integers,  $M'$  and  $N'$  are complementary sets of integers,  $M \neq M'$  and both  $X(M,N)$  and  $X(M',N')$  are included in a set of equations  $Y$ , then  $Y$  is not consistent.

Proof. Let  $m^*$  be an integer that belongs to  $M$  and not to  $M'$ . (The existence of such an integer can be assumed without loss of generality because, since  $M \neq M'$ , there exists an element which is in one and not the other.) Then  $m^*$  is in  $N'$ . Since  $m^*$  is in  $M$ ,  $(2^{m^*}x + x = 2x + x)$  is in  $X(M,N)$ . Since  $m^*$  is in  $N'$ ,  $(2^{m^*}x + x = x)$  is in  $X(M',N')$ . Since both  $X(M,N)$  and  $X(M',N')$  are included in  $Y$ , both of the equations are in  $Y$ . Therefore,  $(2x + x$



$= x \setminus$  and  $\langle 2y + y = y \setminus$  are in  $\text{cl}(Y)$ , the closure of  $Y$ ; and in view of part (1) of 3.2,  $\langle x = y \setminus$  is in  $\text{cl}(Y)$  as well. Therefore,  $Y$  is not consistent. #

Lemma 3.8. There exists at least continuum different equationally complete classes of equations.

Proof. It is well known that there are continuum different couples of complementary sets of integers. By Theorem 1.27, for any given complementary sets  $M$  and  $N$ ,  $1 \in M$ ,  $X(M, N)$  is included in some equationally complete class of equations. But the previous result showed that the closure of the union of complementary sets can not be contained in an equationally complete set of equations. Therefore, there are at least continuum many equationally complete classes of equations. #

Lemma 3.9. There exists at most continuum many equationally complete classes of equations.

Proof. By Definition 1.3 the class of equations is the class of expressions  $\langle \alpha = \beta \setminus$  where  $\alpha$  and  $\beta$  are terms. But there exist countably many terms. Thus the cardinality of the class is at most equal to that of the continuum. #

Theorem 3.10. There exist exactly continuum different equationally complete classes of equations.

Proof. The proof follows directly from Lemma 3.8 and Lemma 3.9. #

Theorem 3.11. There exist continuum different

equationally complete classes of commutative equational algebras.

Proof. This follows from Lemma 3.5, which shows that for any two complementary sets of integers there is a unique commutative algebra with an equationally complete set of identities. #

## CHAPTER IV

### REMARKS ON ALGEBRAS WITH MORE THAN ONE OPERATION

In the previous sections, we have studied only those algebras that have only one operation. In this section we will discuss generalizations of the basic results to algebras with more than one operation. Let  $\ddot{A} = \langle A, F_n \rangle$  be an algebra where  $F_n$  is any sequence of  $n$  finitary operations on  $\ddot{A}$ , say:  $+_1, +_2, \dots, +_n$ . In order to avoid excessive complication of the notation, we shall assume that each operation is a binary one. This will fit most of the examples to be considered. Extending this idea to operations of higher degree should present no conceptual difficulties to the reader.

To extend our work to algebras of this type, we must first add more metamathematical symbols to our language. Instead of only "+" we will have "+<sup>1</sup>", "+<sup>2</sup>",  $\dots$ , "+<sup>n</sup>" as functional variables. This means that the set of identities of  $\ddot{A}$  will contain all the identities for  $\langle A, +_1 \rangle$ ,  $\langle A, +_2 \rangle$ ,  $\dots$ ,  $\langle A, +_n \rangle$  and all the equations that represent the relations between the operations of  $F_n$ ; for example, there might be a distributive law of  $+_1$  over  $+_2$ , et cetera.

The concept of homomorphic image is generalized. We

say that  $g$  is a homomorphism from  $\ddot{A} = \langle A, F_n \rangle$  to  $\ddot{A}' = \langle A, F_n' \rangle$  if for all  $a, b \in A$  and  $+_1 \in F$ ,  $g(a +_1 b) = g(a) +_1 * g(b)$ . The concept of subalgebra is generalized in a similar manner. We also generalize the definition of class of terms to the least class  $T$  of strings such that all variables are in  $T$  and if  $+_1 \in F_n$  and  $\alpha$  and  $\beta \in T$ , then  $(\alpha +_1^1 \beta)^1 \in T$ .

Theorem 1.32 must be revised somewhat. The operation  $+_k$  is defined as  $E(\alpha) +_k E(\beta) = E((\alpha +_k^k \beta)^1)$ . Then the algebra  $\langle T, F_n \rangle$ , where  $F_n = \{+_1, +_2, \dots, +_n\}$ , would be a model of  $X$ , a consistent class of equations.

The next result we will prove is that all Boolean algebras are equationally complete. To do this we will first define a characteristic algebra  $\ddot{C}(BA)$ . Then we will show that  $\ddot{C}(BA)$  generates the class of all Boolean algebras. Then we will show that each Boolean algebra contains a subalgebra that is isomorphic to  $\ddot{C}(BA)$ . From this we will conclude that all Boolean algebras are equationally complete.

Definition 4.1. We will define  $\ddot{C}(BA)$  as the set  $\{0, 1\}$  with the binary operations  $+_1$  and  $+_2$  and the unary operation  $\sim$ , defined as follows:

| $+_1$ | 0 | 1 |
|-------|---|---|
| 0     | 0 | 1 |
| 1     | 1 | 1 |

| $+_2$ | 0 | 1 |
|-------|---|---|
| 0     | 0 | 0 |
| 1     | 0 | 1 |

| $\sim$ |   |
|--------|---|
| 0      | 1 |
| 1      | 0 |

Lemma 4.2. Let  $\ddot{A} = \langle A, +_1, +_2, \sim \rangle$  be a Boolean algebra. Then  $\ddot{A}$  is in the class generated by  $\ddot{C}(BA)$ .

Proof: We want to show that  $\ddot{A}$  is isomorphic to a subalgebra of  $\prod_{a \in A} \ddot{C}(BA)_a$ . The Stone Representation Theorem states that every Boolean algebra is isomorphic to a subalgebra of the power set  $P(S)$  of a set  $S$  with operations of union and intersection. The reader can find a proof of this well known result in Mac Lane and Birkhoff [7]. Then  $\ddot{A}$  is isomorphic to  $\ddot{A}^*$  for some  $\ddot{A}^* = \langle A^*, +_1, +_2, \sim, I, 0 \rangle$  a Boolean subalgebra of  $P(S)$  for some set  $S$ . We define  $f$ , a map from  $A^*$  to  $\prod_{a \in A} \ddot{C}(BA)_a$  as follows: for every  $k \in A^*$  and every  $a \in A$ ,  $f(k)_a = 1$  if  $a \in k$  and  $f(k)_a = 0$  otherwise.

It is obvious that  $f$  is well defined. It is likewise clear that  $f$  is a homomorphism under  $"+_1"$ ,  $"+_2"$ , and  $"\sim"$ . We know that  $f$  is one-to-one since if  $f(k) = f(k^*)$  for some  $k$  and  $k^*$  in  $A^*$ , then for each  $b \in A$ ,  $f(k)_b = f(k^*)_b$ . By the definition of  $f$ , this occurs if and only if  $b \in k$  and  $b \in k^*$ , or  $b \notin k$  and  $b \notin k^*$ . Hence  $k = k^*$ . #

Lemma 4.3. Every Boolean algebra contains a subalgebra isomorphic to  $\ddot{C}(BA)$ .

Proof. Let  $\ddot{A}$  be a Boolean algebra. Let  $a \in A$ . Then  $\tilde{a} \in A$  and  $a +_1 \tilde{a}$  and  $a +_2 \tilde{a}$  are in  $A$ . Therefore  $0$  and  $I$  are in  $A$ . But the set  $\{0, I\}$  is closed under  $"+_1"$ ,  $"+_2"$ ,

and " $\sim$ " and is isomorphic to  $\ddot{C}(BA)$ . #

Theorem 4.4. Every Boolean algebra is equationally complete.

Proof. Follows directly from Lemma 4.2 and Lemma 4.3.

The reader may find other examples of equationally complete algebras in the literature, especially Kalicki and Scott [5] and Kalman [6].

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